

CARBONYL THERAPY

By William Frederick Koch

AS PROFESSOR of Physiology at Wayne University, Detroit (1914-1919), I taught an original interpretation of sugar oxidation which is in conflict with the presently accepted Krebs Cycle theory. Our hypothesis conceived the oxidations of function as the process which ordinarily destroys the toxins of biological origin. On the other hand, disease producing toxins we held to be incompletely burned products of germ metabolism, presenting fairly mobile hydrogen atoms. But these may not be able to reduce the catalysts of the cellular oxidations. To be burned, they may require more active catalysts than those presented by the intermediaries of sugar oxidation. This, we postulate, will depend upon the extent to which the oxidations in the germ are inhibited.

The intermediaries of sugar oxidation we conceive to be produced by dehydrogenation at the carbon atom alpha to the carbonyl group with removal of hydroxyl at the beta carbon atom. Further dehydrogenations and dehydrations and oxygen additions lead to the formation of carbonyl groups in conjugation with ethylene linkages, and ultimately in the production of the suboxides of carbon $O=C=C=O$ and $O=C=C=C=O$. These, in the presence of oxygen, become carbon dioxide, and in its absence, polymerize to form long chains of carbonyl groups that serve

as very active dehydrogenators. Such highly active carbonyl groups are able to thus prepare the toxin molecule for oxygen addition to form a peroxide free radical when oxygen again enters the field.

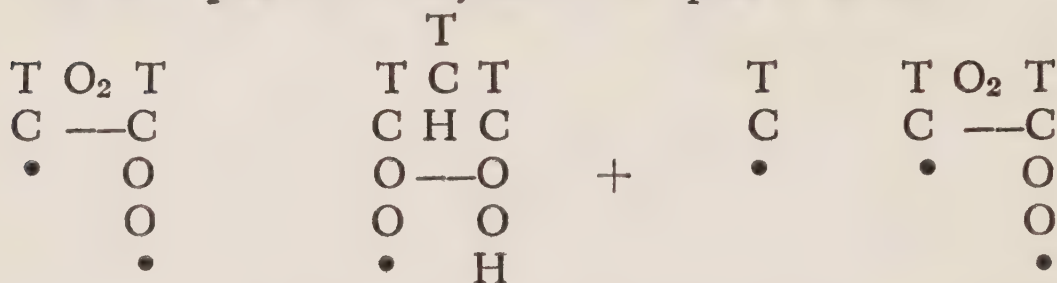
It is our observation that cultures of pus germs taken from the milk of badly infected udders of dairy cattle and incubated 84 hours give a filtrate through a Berkefeldt which is highly toxic, but that after the animal is treated with the catalysts of oxidation presenting highly active carbonyl groups, the pathology is sharply reversed and cultures taken while recovery is in progress and likewise incubated give a filtrate that is non-toxic. We judge from this that the production of the toxin by the germ was due to deficiency in its own oxidations and that the catalyst we offered it restored its oxidations so that the products of its metabolism were completely burned and could not produce pathogenic effects, such as neurosis or abnormal energy transfers through photosensitization.

The toxin in the field undergoes a chain reaction in which the toxin molecule is dehydrogenated to form a free radical and this combines molecular oxygen to form a peroxide free radical, and the latter dehydrogenates the next toxin molecule with which it collides to form a free radical which combines molecular oxygen to form another peroxide free radical. Thus the oxidation chain is perpetuated until all toxin has been converted. It is evident then that the toxin is converted to the carrier catalyst state of structure which serves as antitoxin. The cause is converted into the curative agent. This reaction chain may be pictured as follows:

Let R represent the major part of the remedy molecule and CO its activated carbonyl group. Let T represent the major part of the toxin molecule and H its highly mobile hydrogen atom. $RC^\bullet$ represents the remedy free radical with its unpaired electron, and $TC^\bullet$ the toxin free radical.

Then—

R C O plus T C H yield T C• plus R C OH

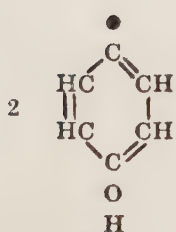
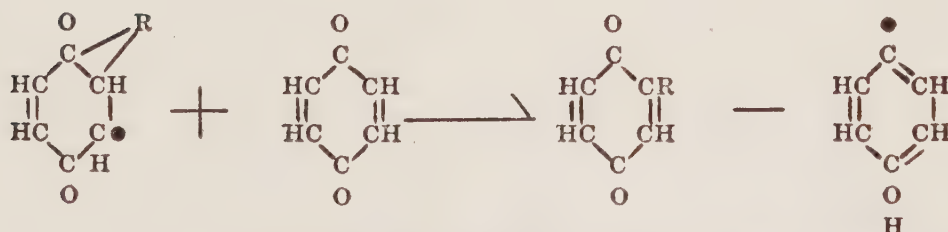
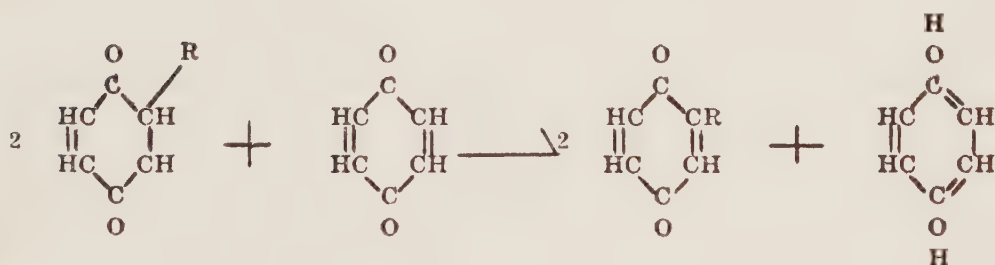
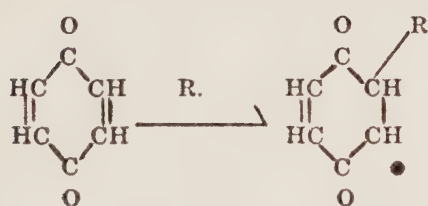


Isolation of any of the very transitory intermediaries of sugar oxidation within living cells is out of the question. They must be hypothesized. The only support we have to our theory is utility of the catalysts employed to treat disease, and the recognition of intermediaries of oxidation of sugar in plants and animals which are modified for functional purposes, and thus stabilized. The keto-steroids of sex gland and adrenal origin illustrate the utilization of the conjugated system of carbonyl and ethylene outlined in our hypothesis. Both the functional and detoxication effects of this atomic system are thus demonstrated. (Many reports on Cortisone Therapy.) (Adair, *Annals of Surgery*, 1946, volume 1023, page 123.) In the intermediaries of sugar oxidation as modified by the mould, the same grouping is observed, and here the oxidations it is able to initiate serve as protective against infection. In the vitamins B, C and K, this same grouping is presented and its ability to detoxicate carcinogens and bacterial toxins through induced oxidation is again proven. (Warren, *Biochem. Journal*, volume 37, p. 338. 1945.) (Geiger and Conn, etc. *J.A.C.S.* Vol. 67. 1945.)

In all of these biological products the adaptation of the molecule to specificity of function necessitated the substitutions in the ethylene linkage which hinder the dehydrogenating action of the carbonyl group although this group is here placed to initiate energy production for the special function. We realized this effect of the substituents long before the keto-steroids and antibiotics or vitamins

were studied in regard to protective powers. We offered benzoquinone as an ideal oxidation initiator through carbonyl activated by conjugation with ethylene. However, this very ethylene linkage may also serve to block the oxidation chain under certain circumstances.

If the oxidation chain is progressing nicely so that toxin molecules are converted to free radical carrier catalysts, the introduction of another molecule of oxidation initiator may not speed up the process, but actually retard it. This is especially true if the initiator carbonyl group is conjugated with ethylene. The process may be outlined thus, — $R^\bullet$ represents the carrier free radical:



yield the quinhydrone. Thus free radicals are inactivated by benzoquinone and the therapeutic action is abolished as clinical tests have proven. We therefore conclude that the therapeutic action of the reaction systems we used must be attributed to the free radicals they contained, and which were responsible for the oxidation and polymerizations in progress in these systems.

Once the carrier free radical is formed, and molecular oxygen is present, the conversion of the substrate toxin to carrier of the oxidation chain will thus progress until all toxin is converted. So long as no inhibitor substance enters the field, this will be true. The success of this therapy then will depend largely on the prevention of inhibitor substances gaining admission.

It is obvious from the foregoing that the oxidation chain is best initiated when the amount of initiator is small as compared to the toxin substrate it acts upon. Each molecule of carbonyl catalyst therefore should be surrounded by a shell of toxin molecules, so that after the first initiative dehydration, the field is occupied by carrier free radicals and toxin substrate which is being converted. Every initiator carbonyl ethylene molecule that enters the field acts as an absorber of the curative carrier free radicals.

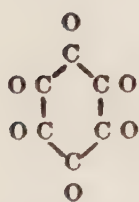
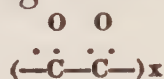
Thus the ideal dosage of the initiator catalyst is very minute, and the 6x to the 12x homeopathic potencies work best in our experience. A dose of one cubic centimeter of the 12x potency carries many trillions of molecules, a quantity that is really excessive. Higher potencies as the 15 and 18 have done even better at times. The matter of dosage selection is an interesting subject but in this brief space we can no more than illustrate that in Nature the most striking physiological action has recently been proven for such solutions as the thirtieth potency (10-30) concentration. Thus but one molecule of Crocetin per cell has been proven by Kuhn and Mowus (1938-1940), (Fieser Text Organic Chem. p 1001, p 753. 1944 edition) and others to activate sex differentiation in algae, (genus *Chlamydomonas*) and by Professor Gilbert Smith in dilutions of one part to 250,000,000,000,000 parts of water to bring on the sex development of algae and the development of the combs of capons. (Science Illustrated February 1947.) Swedish scientists have confirmed this observation.

It should be noted too that biological substances of such high activity in extreme dilution possess the carbonyl

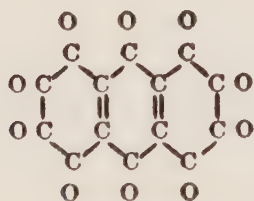
group in conjugation with ethylene. They thus serve as oxidation initiators.

The presence of the methyl group as substituent to the conjugated ethylene groups makes the availability of the PI electrons far greater there and hence the migrations of electrons to the carbonyl groups are increased tremendously. The high activity of carbonyl is made possible in this way in Echinochrome A, which is secreted by Sea Urchin ova, and thus the activation of oxidation is secured in the sperm to give it energy to travel to the egg and fertilize it. Here again the energy producing oxidations for function are illustrated. We may offer many more instances of support to our theory of sugar oxidation, and its application to the boosting of the natural Immunity against disease.

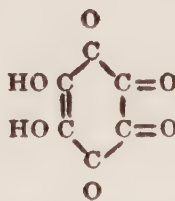
While benzoquinone in the 6x dilution has proven effective in certain toxic states, infections and even in some cases of cancer, its limitations have made the use of the serial systems of carbonyl groups with free radical terminals preferable. The molecules we use also include Rhodozonic acid, Triquinoyl, Malonide and the long chains of carbonyl groups representable by the formula



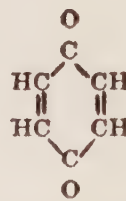
Triquinoyl



Malonide



Rhodozonic Acid



Benzoquinone

The field of action we find to be general, including the pathogenic viruses and bacteria, as well as carcinogenic poisons of industrial and biological origin. Brucellosis, Streptococcus and Staphylococcus infections of the udder have been investigated by the Department of Agriculture of the Province of British Columbia, Canada, over a period of five years. They report data showing prompt curative results of over 85% in several series of cases numbering in the

thousands. (Annual Reports of the Minister of Agriculture to Parliament. 1944, 45, 46, 47, 48, 49.) Prompt disappearance of pathogenic germs and dissolution of fibrosis with restoration of normal milk production has been the rule. Reports of the cure of Brucellosis without change in the agglutinin test, but with restoration of health and fertility and normal placenta and calf at normal term has been the rule. Johne's disease (paratuberculosis) is also reported cured in equal percentage.

Of the metabolic diseases, the most interesting is ketosis. This disease of cattle is a defect in the oxidation of sugar and presents itself as a perfect test of our hypothesis. Where the disease was well established and no expectancy for recovery was had by the veterinarians and owners, only one dose of the catalysts brought about recovery in some ninety per cent of the cases, and in some series, practically one hundred per cent, were reported permanently cured. Another confirmation of our hypothesis came with the cure of perverted and hindered function in dairy cattle. The special function concerned and in which the farmer is most interested is fertility of the animals. Infertility due to perverted ovarian function and to ovarian cysts which had persisted for many months and sometimes for a year or more before treatment were reported normal in a month or two after treatment and that the recoveries were permanent in some 80 per cent of the cases which were otherwise incurable.

Among the virus diseases, swamp fever in horses, a 100 per cent fatal disease, was first investigated by Dr. Ried in an insect infested district of Texas. Several series, totaling some two hundred animals, were treated and the recovery percentage was reported as from 85 to 96 per cent in different groups. They included the latest as well as the earliest stages of the disease. Further virus diseases in animals were studied under the courtesy of the Minister of Agriculture of Brazil. Several hundred cows, pigs and dogs were treated for Hoof and Mouth disease, distemper and

some rare types of infection. The official reported cures run from 80 per cent to 96 per cent when all stages were treated, even though in some instances the animals could not raise themselves from the ground.

The notable feature among all cases treated where interferences were successfully eliminated, was that the recovery process gave evidence of its activity within six hours after the administration of the treatment, in the acute cases. The same observation was made in humans treated for Infantile Paralysis and the recovery percentage ran equally high or higher. At least we have no knowledge of any failures yet, even though some cases with respiratory paralysis and impending death were in the list and these recovered most satisfactorily.

In the acute cases of Infantile Paralysis where the treatment was given within a day or two of the establishment of paralysis, the recovery is found to be complete with neither paralysis or atrophy remaining. When treated later than two days after paralysis is established, a slight amount of paralysis or atrophy may remain, and the extent depends upon the length of time that intervened in which the virus has a chance to actually kill the nerve cells of the spinal cord. However, it is also observed that in the chronic cases where the atrophy and paralysis of the legs was quite complete, even for years, and sometimes as long as twenty years, the recovery process may approach 90 per cent of completion, with restoration of ability to walk without braces, etc., and in the restoration of the muscles.

These facts confirm our original hypothesis in that it is seen that function is hindered by the virus which is still present in the nerve cells, and that the hindering factor is an incompletely oxidized product of metabolism in the virus. It inactivates the catalyst of oxidation for function in the cell so that while the cell is not dead, and can be restored to function, it still is not functioning and paralysis is maintained until a powerful carrier or initiator catalyst is supplied to restore the oxidation process within the virus and

the affected nerve cell. Then the virus no longer produces an inhibitor but burns its products to completion. The incapacitated cells likewise are supplied a means of conducting their oxidations of function, and so the paralysis disappears and the muscles develop again to the extent that they have restored nerve function. And this, in some cases of complete paralysis, has been about 85 per cent, or 90 per cent in very chronic cases.

Cancer is likewise a virus disease, as is well proven today. In the Bulletins of the Koch Cancer Foundation of 1925-1928 we discussed the chemical properties of the toxin of this infection and indicated the ultra-microscopic nature of the agent. Now the electron microscope proves the correctness of this publication. The importance of the virus nature of the serious acute infections in animals and man and of the several so-called incurable chronic diseases as cancer, is that it shows the avenue for successful attack.

The first official cases of cancer treated, were far advanced and classified as utterly hopeless of cure. They were cases of generalized cancer involving the vital internal organs. Seven such cases were chosen as test material by a special committee appointed by the Wayne County Medical Society in 1919. Three weeks were allowed for the treatment to show its results. At the end of this period, the patients were sent home and the investigation was closed. Of the seven cases treated, at least three made recoveries. Maybe more did, but they were not followed since they were sent home to distant cities and states where examination and follow-up care was not possible. Of the three recoveries, one was found cured over five years afterward and another was still cured fifteen years afterward and was killed by accident, autopsy revealing no cancer as a cause of death, but brain hemorrhage and skull fracture. The other lived in another state and could not be followed over two years. This investigation was used as a basis to officially falsify, and though the officials who condemned have full knowledge of the facts, they still condemn this 43 per cent

recovery rate as a failure and the treatment as a fraud. Yet of the 7,000,000 Americans who have died of cancer since 1919, according to this officially proven curative percentage, some 3,000,000 persons would have been put back to work in good health, winning bread for their families, even if they had been treated in the far advanced stages of the disease.

Cancer is well recognized today by leading experts as a disease of defective oxidation. In fact, the mechanism that conducts the oxidations of function is no longer present. The cancer cell is thus described as *Anaplastic*. It resembles in this respect the anterior horn cells of Infantile Paralysis which are not able to burn the anti-oxidant inhibitors produced by the virus which keeps the cell paralyzed. The introduction of the appropriate initiator and carrier catalysts of oxidation have been followed by recovery in like manner in both diseases. In cancer, however, an excess of cells that have no utility and which are much out of order are disposed of like so much clotted blood. The process is technically named *organization*. Calcification is the first step of note, shown in microscopic sections of cancer undergoing cure. Next, the cells undergo liquefaction and small blood capillaries grow into the areas as fast as they liquefy. The growth is thus replaced by blood vessels and this is followed by ingrowth of normal tissue cells which complete the reconstruction of destroyed tissue along normal physiological functioning lines. It is not surprising then that many cases of far advanced completely proven cancer of the uterus have been able to give birth to normal children four or five times and still remain cured as long as thirty years after treatment. Reconstruction of the stomach and other organs with the cure of the disease is often reported. The process of recovery from its inception will be favored or hindered by attendant circumstances in the diet, inhibitor substances in the air, such as exhaust fumes from internal combustion engines, tar products and dietary articles that we can pretty well classify for each person.

Cancer must be classified as an allergy in our hypothesis. The mechanism is not a mystery at all, but can be readily understandable. The toxin has two properties. It is an oxidation inhibitor through the highly mobile hydrogen atom attached to the carbon atom in alpha position to the most active carbon to carbon or nitrogen to nitrogen double bond. The other action is the fluorescent photosensitization of the mitotic mechanism in the cell through energy transfer from hydrolytic glycolysis to the chemical processes of this mechanism. This energy is transferred by the double bonds just mentioned. It is continuous so long as the cell lives and forces it to multiply over and over again each daughter cell carrying on the same process. The carcinogen is active in exceedingly high dilution as are other allergenic agents.

So far as we are able to make tests, the principle is proving to hold. Leprosy, tuberculosis of far advanced stages, undulant fever, rheumatic fever, acute and chronic, and the various acute and chronic infections met in the United States and in the Tropics, have all responded in like percentage regardless of the name or symptom picture of the disease. These diseases are all correctible in the same way also, and so the principle may be considered as true until proven otherwise, that disease producing germs are suffering in various degrees from a crippling of their oxidation mechanism and this effect is handed over to the cells they invade. They multiply at top speed to survive their defect, and when the defect is corrected by supplying an appropriate oxidation catalyst, they carry their oxidations to completion and no longer produce toxins. They do not multiply rapidly any more either, and so they disappear as toxin producers and remain only so long as there is dead tissue debris to be changed into material satisfactory for food for plants. They thus return to their normal place in the economy of Nature. The tissue cells that were affected are able to normalize in the same way and return to carrying on effective oxidations for function.

Naturally proving such a hypothesis is rather upsetting to the smug routine of the practice of medicine, but many of the wise older physicians and the energetic younger men see the advantage in practicing in line with this newer philosophy, and a brighter day is dawning for the health of mankind.